

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

File Transfer Protocol (FTP)

Definition

File Transfer Protocol (FTP) is the standard TCP/IP application-layer protocol used to transfer files from one host to another.

Why FTP?

- Transfers files between different computers.
- Handles different file naming conventions.
- Supports different data representations.
- Supports different directory structures.
- Better than HTTP for transferring large files and files of different formats.

FTP Components

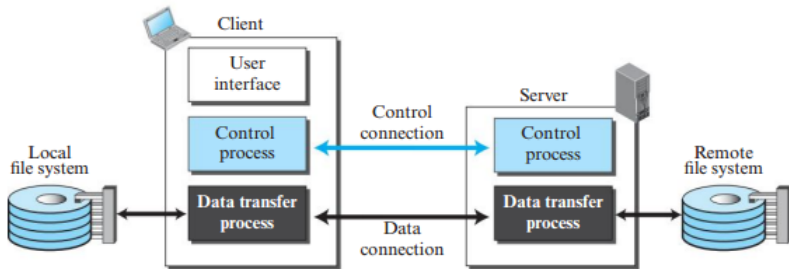
Client

- User Interface
- Client Control Process
- Client Data Transfer Process

Server

- Server Control Process
- Server Data Transfer Process

Figure 2.17 *FTP*



FTP Connections

FTP uses two separate TCP connections.

Control Connection

- Transfers commands and responses.
- Connects Client Control Process and Server Control Process.
- Uses TCP Port 21.

Data Connection

- Transfers actual files.
- Connects Data Transfer Processes.
- Uses TCP Port 20.

Why Two Connections?

Advantages

- Commands and data are separated.
- Control connection transfers only one command/response at a time.
- Data connection efficiently transfers files.
- Improves FTP performance.

Exam Point

FTP uses separate Control and Data connections, making it more efficient than combining both into a single connection.

Lifetime of FTP Connections

Control Connection

- Opened once when FTP session starts.
- Remains active throughout the session.
- Closed only after user logs out.

Data Connection

- Opened whenever a file transfer begins.
- Closed immediately after the transfer finishes.
- Opened repeatedly for multiple files.

Characteristics

- Uses NVT ASCII format.
- Similar communication style as TELNET.
- Commands are sent from Client to Server.
- Responses are sent from Server to Client.
- Every command line ends with CR + LF.

Common FTP Commands (Part 1)

Command	Purpose
USER	User identification
PASS	Password authentication
PWD	Display current directory
CWD	Change directory
CDUP	Move to parent directory
LIST	Display files/directories
MKD	Create directory
RMD	Remove directory
QUIT	Logout from FTP session

Common FTP Commands (Part 2)

Command	Purpose
RETR	Download file from server
STOR	Upload file to server
DELE	Delete a file
RNFR	Old filename for rename
RNTO	New filename for rename
PORT	Client selects data port
PASV	Server selects data port
ABOR	Abort current command
TYPE	Select file type
STRU	Select data structure
MODE	Select transmission mode

Every FTP command generates at least one response.

Response Format

- Three-digit numeric code
- Followed by explanatory text

Meaning of digits

- First digit : Status of command
- Second digit : Functional area
- Third digit : Additional information

Common FTP Response Codes

Code	Meaning
125	Data connection open
150	File status OK
200	Command OK
220	Service Ready
221	Service Closing
225	Data connection open
226	Closing data connection
230	User login successful
250	Requested file action OK
331	Username OK, password required
425	Cannot open data connection
450	File unavailable
452	Insufficient storage
500	Syntax error
501	Invalid parameters
530	User not logged in

FTP Data Connection

Unlike the control connection, the data connection is created only when required.

Connection Establishment

- 1 Client opens an ephemeral port.
- 2 Client sends the PORT command.
- 3 Server performs an active open using Port 20.

Exam Point

FTP Control Connection uses Port 21.

FTP Data Connection uses Port 20.

Communication over Data Connection

Before transferring a file, FTP defines three communication attributes.

- 1 File Type
- 2 Data Structure
- 3 Transmission Mode



These attributes solve interoperability problems between different operating systems.

FTP supports three data organizations.

- **File Structure (Default)**
 - Continuous stream of bytes
- **Record Structure**
 - File divided into records
- **Page Structure**
 - File divided into numbered pages
 - Random or sequential access

FTP File Types and Transmission Modes

File Types

- ASCII
- EBCDIC
- Image (Binary)

Transmission Modes

- Stream Mode (Default)
- Block Mode
- Compressed Mode



FTP File Transfer

FTP supports three transfer operations.

- Retrieve File (Server → Client)
- Store File (Client → Server)
- Directory Listing (Server → Client)

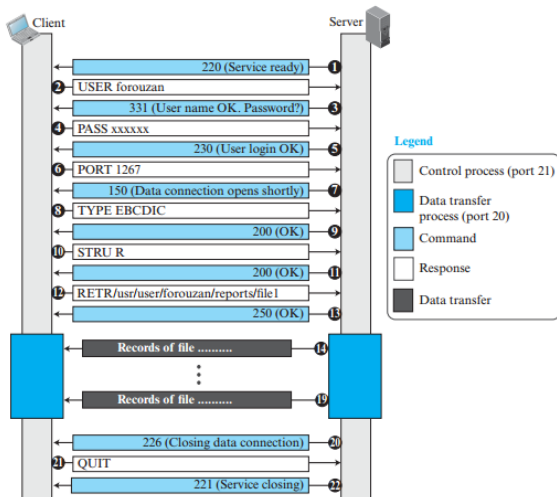
File transfer occurs over the Data Connection under the control of commands sent through the Control Connection.

Example 2.11 : Retrieving a File

The example illustrates FTP retrieval of a file.

- Control connection remains open.
- Data connection opens for file transfer.
- File transferred in multiple sections.
- Data connection closes after transfer.
- QUIT command terminates the FTP session.

Figure 2.18 Example 2.11



Example 2.12 : FTP Session

Typical FTP Session

- Connect to FTP server
- USER and PASS authentication
- Login successful
- Directory listing using LIST
- Data transferred
- QUIT command ends session

Observation

Server responses (220, 331, 230, 150, 226, 221) indicate the progress of the FTP session.

FTP Security

Problem

FTP transfers usernames, passwords, and files in plaintext.

Risks

- Password interception
- File interception
- Unauthorized access



Solution

- SSL/TLS can be added to FTP.
- Secure version is called **SSL-FTP (FTPS)**.
- Secure file transfer can also be achieved using SSH-based protocols.

Key Takeaway

Standard FTP is not secure because it does not encrypt data.